

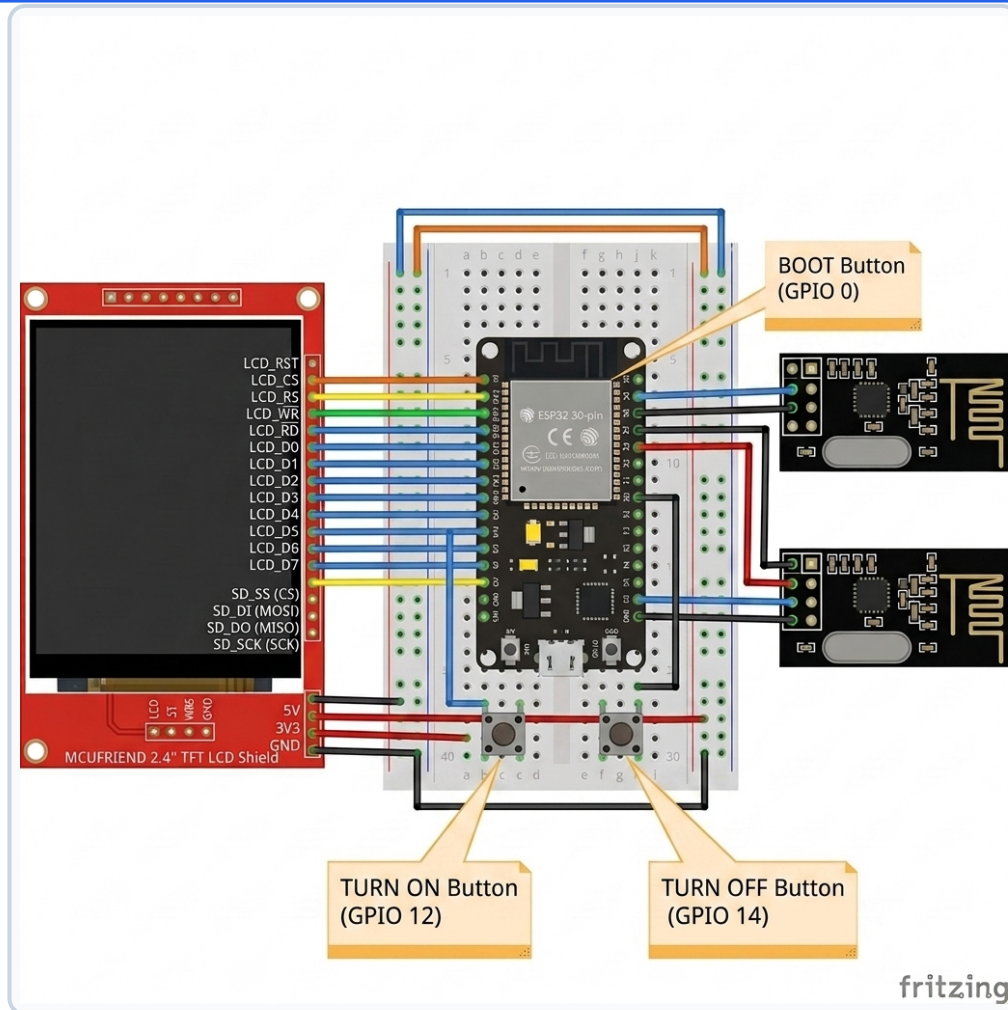
ESP32 BT JAMMER & WI-FI SECURITY SUITE

Master Hardware Pinout, RF Transceiver & Wiring Diagram Guide

LEGAL & EDUCATIONAL DISCLAIMER:

This device design and documentation are strictly for educational and authorized penetration testing in controlled environments. Operating RF jammers or packet injection without authorization is illegal.

1. Master Hardware Breadboard Wiring Diagram



COMPONENTS SHOWN IN WIRING DIAGRAM:

- **ESP32 30-Pin Dev Board:** Center of breadboard (30-pin DIP layout)
- **MCUFRIEND 2.4" TFT Shield:** Left side — 8-bit parallel bus + SPI
- **NRF24L01 #1 (PCB Ant.):** Right top — CE: GPIO 22, CSN: GPIO 21
- **NRF24L01 #2 (PCB Ant.):** Right bottom — CE: GPIO 16, CSN: GPIO 15
- **TURN ON Pushbutton:** Lower left — GPIO 12 & GND (INPUT_PULLUP)
- **TURN OFF Pushbutton:** Lower right — GPIO 14 & GND (INPUT_PULLUP)
- **BOOT Button (GPIO 0):** Built-in on ESP32 board (BACK navigation)
- **Status LED (GPIO 2):** Built-in on ESP32 board (Active indicator)

1B. Breadboard Wire-by-Wire Connection Reference

Wire Color Key:

RED 3.3V / 5V Power Supply Rail	BLACK GND Ground Rail	BLUE SPI SCK (Clock) — GPIO 18
PURPLE SPI MOSI (Data Out) — GPIO 23	CYAN SPI MISO (Data In) — GPIO 19	ORANGE NRF CE / CSN control lines
GREEN TURN ON Button wire — GPIO 12	YELLOW TFT control lines (RS, CS, RST, WR)	VIOLET TURN OFF Button wire — GPIO 14

Complete Breadboard Connection Table — All Wires

From Component / Pin	ESP32 GPIO	Wire Color	Destination / Rail
Breadboard +Rail (3.3V)	3.3V pin	RED	! NRF24 #1 VCC, NRF24 #2 VCC
Breadboard " Rail (GND)	GND pin	BLACK	! NRF24 #1 GND, NRF24 #2 GND, BTN GND legs
TFT Shield 5V/3V3	5V / 3.3V	RED	! Breadboard power rail
TFT Shield GND	GND	BLACK	! Breadboard GND rail
SPI Clock (SCK)	GPIO 18	BLUE	! NRF24 #1 SCK, NRF24 #2 SCK, TFT SD_SCK
SPI MOSI	GPIO 23	PURPLE	! NRF24 #1 MOSI, NRF24 #2 MOSI, TFT SD_DI
SPI MISO	GPIO 19	CYAN	! NRF24 #1 MISO, NRF24 #2 MISO, TFT SD_DO
NRF24 #1 CE	GPIO 22	ORANGE	! NRF24 #1 CE pin
NRF24 #1 CSN	GPIO 21	ORANGE	! NRF24 #1 CSN pin
NRF24 #2 CE	GPIO 16	ORANGE	! NRF24 #2 CE pin
NRF24 #2 CSN	GPIO 15	ORANGE	! NRF24 #2 CSN pin
TFT LCD_RST	GPIO 33	YELLOW	! TFT Shield LCD_RST pin
TFT LCD_CS	GPIO 5	YELLOW	! TFT Shield LCD_CS pin
TFT LCD_RS (DC)	GPIO 4	YELLOW	! TFT Shield LCD_RS pin
TFT LCD_WR	GPIO 2	YELLOW	! TFT Shield LCD_WR pin
TFT LCD_RD	3.3V Rail	RED	! Tie HIGH (no data read needed)
TFT LCD_D0	GPIO 12	BLUE	! TFT Shield D0 (shared w/ TURN ON btn)
TFT LCD_D1	GPIO 13	BLUE	! TFT Shield D1 (shared w/ SD_SS)
TFT LCD_D2	GPIO 26	BLUE	! TFT Shield D2
TFT LCD_D3	GPIO 25	BLUE	! TFT Shield D3
TFT LCD_D4	GPIO 17	BLUE	! TFT Shield D4
TFT LCD_D5	GPIO 27	BLUE	! TFT Shield D5
TFT LCD_D6	GPIO 14	BLUE	! TFT Shield D6 (shared w/ TURN OFF btn)
TFT LCD_D7	GPIO 32	BLUE	! TFT Shield D7
SD Card SS (CS)	GPIO 13	YELLOW	! TFT Shield SD_SS
TURN ON Pushbutton	GPIO 12	GREEN	Leg 1 ! GPIO 12, Leg 2 ! GND Rail
TURN OFF Pushbutton	GPIO 14	VIOLET	Leg 1 ! GPIO 14, Leg 2 ! GND Rail
BOOT Button	GPIO 0	—	Built-in on ESP32 board (no wire needed)
Status LED	GPIO 2	—	Built-in on ESP32 board (no wire needed)

2. ESP32 Pin Allocation Overview

ESP32 30-PIN DEV BOARD PINOUT			
Radio 1 CE	GPIO 22	GPIO 21	Radio 1 CSN
Radio 2 CE	GPIO 16	GPIO 15	Radio 2 CSN
Touch T_CS	GPIO 27	GPIO 4	TFT DC / RS
TFT Backlight	GPIO 32	GPIO 5	TFT CS
TFT Reset	GPIO 33	GPIO 18	SPI SCK (Clock)
Physical ON Btn	GPIO 12	GPIO 19	SPI MISO
Physical OFF Btn	GPIO 14	GPIO 23	SPI MOSI
Power Rail	3.3V	GND	Ground Rail

3. Hardware Interfaces (VSPI & ESP32 Wi-Fi Radio)

Signal / Interface	ESP32 Pin	Connected Modules / Mode	Type
SCK (Clock)	GPIO 18	Radio 1, Radio 2, TFT SCK, Touch T_CLK	Shared SPI
MISO (Master In)	GPIO 19	Radio 1, Radio 2, TFT SDO, Touch T_DO	Shared SPI
MOSI (Master Out)	GPIO 23	Radio 1, Radio 2, TFT SDI, Touch T_DIN	Shared SPI
Wi-Fi 802.11 b/g/n	Internal RF	AP Scanning, Promiscuous Sniffer, Web Portal	Internal RF

4. Radio Transceiver Modules (NRF24L01)

Radio #1 (Lower Channels 2402–2440 MHz):

NRF24 #1 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Power Rail (+3.3V Max)	Power / Ground
CE	GPIO 22	Chip Enable	Dedicated Control
CSN	GPIO 21	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	GPIO 18 / 23 / 19	Hardware SPI Bus	Shared SPI

Radio #2 (Upper Channels 2441–2480 MHz):

NRF24 #2 Pin	ESP32 Pin	Function	Signal Category
VCC / GND	3.3V / GND	Power Rail (+3.3V Max)	Power / Ground
CE	GPIO 16	Chip Enable	Dedicated Control
CSN	GPIO 15	SPI Chip Select	Dedicated SPI CS
SCK / MOSI / MISO	GPIO 18 / 23 / 19	Hardware SPI Bus	Shared SPI

5. MCUFRIEND 2.4" TFT LCD Shield (8-Bit Parallel)

Shield Pin	ESP32 Pin	Function	Signal Category
5V / 3V3 / GND	5V / 3.3V / GND	Display Power & Ground	Power / Ground
LCD_RST	GPIO 33	Hardware Reset	Dedicated Control
LCD_CS	GPIO 5	Chip Select	Dedicated Control
LCD_RS	GPIO 4	Register / Command Select	Dedicated Control
LCD_WR	GPIO 2	Write Enable	Dedicated Control
LCD_RD	3.3V (Tie HIGH)	Read Enable (Not Used)	Power
LCD_D0	GPIO 12	Data Bit 0	8-Bit Parallel
LCD_D1	GPIO 13	Data Bit 1	8-Bit Parallel
LCD_D2	GPIO 26	Data Bit 2	8-Bit Parallel
LCD_D3	GPIO 25	Data Bit 3	8-Bit Parallel
LCD_D4	GPIO 17	Data Bit 4	8-Bit Parallel
LCD_D5	GPIO 27	Data Bit 5	8-Bit Parallel
LCD_D6	GPIO 14	Data Bit 6	8-Bit Parallel
LCD_D7	GPIO 32	Data Bit 7	8-Bit Parallel
SD_SS (CS)	GPIO 13	SD Card Chip Select	SPI CS
SD_DI (MOSI)	GPIO 23	SD SPI Data In	Shared SPI
SD_DO (MISO)	GPIO 19	SD SPI Data Out	Shared SPI
SD_SCK (SCK)	GPIO 18	SD SPI Clock	Shared SPI

6. Hardware Pushbuttons & System Controls

Button / LED	ESP32 Pin	Wiring Connection	Mode
TURN ON Button	GPIO 12	One leg to GPIO 12, other leg to GND	INPUT_PULLUP
TURN OFF Button	GPIO 14	One leg to GPIO 14, other leg to GND	INPUT_PULLUP
BOOT Button	GPIO 0	Built-in BOOT pushbutton on ESP32 board	Internal (BACK Key)
Status LED	GPIO 2	Built-in LED on ESP32 board	OUTPUT (Active Status)

7. Power Supply & Capacitor Recommendations

STANDARD PCB ANTENNA MODULES (RECOMMENDED):

Draw ~12mA each. Can be powered directly from ESP32 3.3V rail without external capacitors. No decoupling required for typical breadboard setups.

HIGH-POWER PA+LNA MODULES (EXTERNAL ANTENNA):

Draw up to ~115mA peak each. Requires 10µF to 47µF electrolytic or ceramic capacitor soldered directly across VCC and GND pins of each NRF module to prevent brownouts.

HARDWARE ASSEMBLY CHECKLIST

- ' Common Ground (GND): Ensure all grounds (ESP32, TFT, NRF24 #1, NRF24 #2, Buttons) are tied together.
- ' Power Supply: Use a dedicated 5V 2A USB power supply or battery pack during dual radio transmission.
- ' SD Card Format: FAT32 with /wordlist.txt in root directory for offline Wi-Fi password auditing.
- ' Touchscreen Calibration: Use MCUFRIEND TouchScreen library XP=8, YP=A3, XM=A2, YM=9 (300 ohm).